

The market value of variable renewables

Dispatch and variable generation

Reading. *Energy System Models*, sections 2 and 3 (appendix A for the duality machinery); Hirth (2013). Optional: Joskow (2011), López Prol et al. (2020).

Part 1 · Mechanism

35 min

The value of wind in a two-hour system

Two hours of equal length, a calm one and a windy one, with nothing linking them. Three plants:

	capacity, GW	marginal cost, EUR/MWh	availability (calm, windy)
Wind	K	0	0, 1
Coal	2	40	1, 1
Gas	2	80	1, 1
Load, GW	calm 1.5, windy 3		

1. Set up the cost-minimising dispatch of the two hours and derive its optimality conditions. From them, determine the price in each hour for every $K \geq 0$, and mark the values of K at which the price-setting plant changes.
2. Derive wind's capture price $\bar{p}_w(K)$, the base price $\bar{p}(K)$ and wind's value factor $VF_w(K)$ for every $K \geq 0$. Sketch $VF_w(K)$ and name the event behind each kink.
3. Hirth (2013, §3.1) defines the base price as the time-weighted average price; the note's §3.4 weights by consumption. Compute both value factors at $K = 0.5$, and derive the general condition under which Hirth's is the larger of the two.
4. Hirth (2013, §4.3) sorts models by how they treat the capital stock, short, mid and long term, and states that under perfect markets the market value of a variable technology equals the marginal cost saving it brings to the system. Which of his three perspectives is this exercise? Then, for a general system of T hours with a fixed fleet, show that the derivative of minimised system cost with respect to K equals $-\sum_h \gamma_h \lambda_h$ wherever wind is not curtailed, and conclude that wind's capture price is exactly the cost saving per MWh of wind.
5. Still in the general system: show that $\bar{p}_w$ is non-increasing in K , and give the exact condition on the hourly profiles under which $VF_w > 1$ as $K \rightarrow 0$. Hirth calls the second the correlation effect; say what is being correlated with what.

Stop point

The function $VF_w(K)$ with its kinks named, and the derivative of question 4.

If you finish early

Is VF_w itself necessarily non-increasing in K ? Prove it, or construct a two-hour system in which it rises.

The cannibalisation curve, and what drives it

What Hirth finds in market data. Hirth (2013, §3) computes yearly wind and solar value factors from day-ahead prices and hourly generation in Germany and five other countries, 2001 to 2010, and regresses them on the technology's market share (his eq. 4 and Table 5). The value factor starts near one at a share of zero and falls by 1.62 points per percentage point of wind share in thermal systems, by 0.26 in hydro systems. His model of the European market (§5) extends the curve beyond the shares in the data: wind at 0.5 at a 30% share in the mid term and 0.65 in the long term, solar below 0.5 at 15%.

What the model simulates. The note's model of section 3 does the same experiment on a system we know in full: the ten-plant fleet of the note's Table 1, dispatched hour by hour against DK1's 2024 load, wind and solar profiles (every eighth hour of the year, so that each solve takes seconds). The fleet is fixed, nothing is stored or exported, and the market share of a technology is changed by multiplying its installed capacity by a factor s . Three experiments were run:

- wind capacity scaled by s , everything else as it is;
- solar capacity scaled by s , everything else as it is;
- solar capacity scaled by s with wind fixed, and *wind's* statistics recorded, to see what one technology's growth does to the other.

The notebook *E3-notebook.ipynb* reproduces them; running it is optional.

How to read Table 1. One row per solve. The first two blocks report the scaled technology's own statistics: its share of consumption; its capture price $\bar{p}_g$; the base price $\bar{p}$, consumption-weighted as in the note; the time-average price $\langle \lambda \rangle$, which is the base price as Hirth defines it; the value factor $VF_g = \bar{p}_g / \bar{p}$; and curtailment, the share of the energy the weather offered that the system could not use. The third block reports wind's statistics while solar is scaled: the share column is then *solar's* share, the prices and value factor are *wind's*, and the last column is wind's revenue over the sampled year. Figure 1 plots the same numbers: (a) capture price against the technology's own share, with the base price along the same sweep dashed, so that the gap between the two lines is what the value factor measures; (b) the value factor, with curtailment dotted on the right axis; (c) the third experiment, wind's value factor and wind's revenue as solar's share rises.

scale s	share %	$\bar{p}_g$	$\bar{p}$	$\langle \lambda \rangle$	VF_g	curtailed %
<i>Wind scaled: wind's statistics</i>						
0.1	3	72.6	71.9	70.8	1.01	0
0.25	8	68.1	69.0	68.3	0.99	0
0.5	16	64.2	66.3	65.7	0.97	0
0.75	25	59.1	63.0	62.1	0.94	0
1	33	49.2	56.8	56.2	0.87	0
1.5	49	35.7	47.0	46.7	0.76	1
2	59	22.5	36.7	37.1	0.61	9
3	70	13.5	26.6	27.1	0.51	29
4	75	10.1	21.5	21.9	0.47	43
6	81	6.9	15.5	15.8	0.45	59
<i>Solar scaled: solar's statistics</i>						
0.1	0	61.4	57.8	57.2	1.06	0
0.25	1	60.7	57.5	57.0	1.06	0
0.5	1	59.9	57.2	56.7	1.05	0
0.75	2	59.3	57.0	56.4	1.04	0
1	2	58.7	56.8	56.2	1.03	0
1.5	4	56.4	56.1	55.5	1.01	0
2	5	53.8	55.3	54.6	0.97	0
3	7	49.5	53.8	53.2	0.92	0
4	10	44.7	52.2	51.6	0.86	0
6	15	36.3	49.2	48.6	0.74	0
scale s	solar share %	$\bar{p}_w$	$\bar{p}$	$\langle \lambda \rangle$	VF_w	wind revenue, M€
<i>Solar scaled, wind fixed: wind's statistics</i>						
1	2	49.2	56.8	56.2	0.87	49
2	5	47.8	55.3	54.6	0.87	48
4	10	45.6	52.2	51.6	0.87	45
8	20	42.2	46.2	45.6	0.91	41
16	32	43.1	38.4	37.9	1.12	36
24	37	43.5	35.3	34.9	1.23	33

Table 1: The three experiments. Shares are of consumption; prices in EUR/MWh; curtailment as a share of the energy the weather offered; revenue in million EUR over the sampled year.

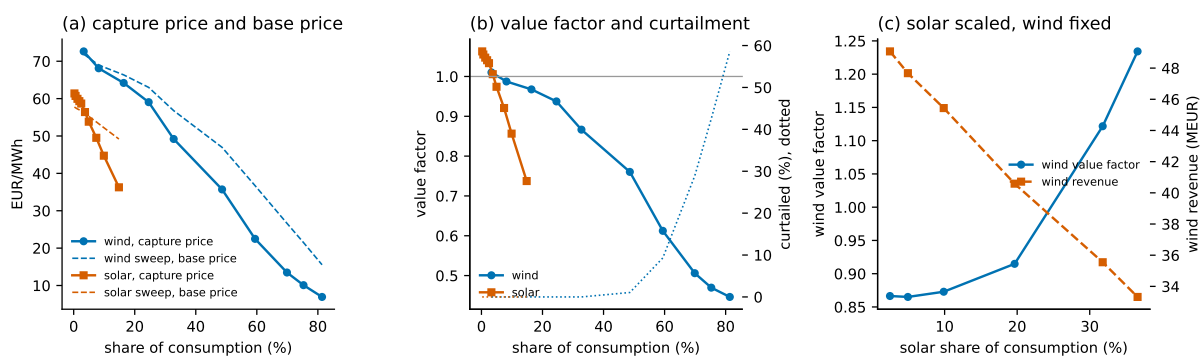


Figure 1: The same numbers. (a) capture price against the technology's own share, base price along the same sweep dashed; (b) value factor, with curtailment dotted on the right axis; (c) solar scaled while wind is fixed: wind's value factor and wind's revenue.

1. From the table, compute the slope of wind's value factor per percentage point of share over the range 3–33% and over 33–59%. Set both against Hirth's regression estimates. Give two reasons his numbers and the model's should differ, and say which way each pushes.

- Between $s = 0.5$ and $s = 1$ the value factor falls with no curtailment; beyond $s = 2$ it falls while curtailment rises. Name the price-setting plant in the hours that matter in each regime, and show that the two regimes are one mechanism. Hirth (2013, Fig. 11) plots the price-setting technology by share of hours and the share of wind sold at a zero price: what would that figure look like for this fleet at $s = 4$? Curtailment also removes energy: say what it does to wind's capacity factor and to its revenue per MW, on top of what it does to the price.
- Solar's value factor falls several times faster per percentage point than wind's. Compute the ratio from the table, then explain it from the profiles, naming the statistic in the note's Table 2 that measures the cause. Hirth's conclusion offers a one-line reason; check that it is the same one.
- Hirth's Table 8 and his Fig. 30 say that every driver he tests works through the level of the base price or the slope of the merit-order curve. In the third sweep solar rises while wind is fixed. Decompose the change in wind's value factor from $s = 1$ to $s = 24$ as $\Delta \ln VF_w = \Delta \ln \bar{p}_w - \Delta \ln \bar{p}$, and say which of the two terms López Prol et al. (2020) would call absolute and which relative cannibalisation. Which number does a wind investor read, and which does a policy maker who indexes support to the value factor read?

Stop point

Two slopes for wind, one ratio for solar against wind, and the two terms of the decomposition.

If you finish early

Hirth (2013, §5.5) finds the effect of a carbon price on wind's value factor ambiguous, through three channels. Which of the three exist in this model? Using the marginal costs of the note's Table 1 at $\tau = 85$ EUR/t, predict the sign for wind at $s = 2$. The notebook lets you check.

Optimal deployment, and whether the drop is a market failure

Hirth (2013, Fig. 3) draws a rising long-run marginal cost of wind (its levelised cost, LCOE) against its falling market value, and calls the intersection the optimal amount of wind. He also writes that profile, balancing and grid-related costs “are not market failures, but represent the intrinsic lower value of electricity during times of high supply, at remote sites, and the economic costs of uncertainty”. The note's §3.5 takes a position on the same question.

- Derive Hirth's figure from the model. Let wind cost c^l per MW and year, annualised, and let $\Gamma = \sum_h \gamma_h$ be its full-load hours. Write the free-entry condition for wind in terms of the capture price of part 1 and show that it is the planner's first-order condition for K (the note's Result 6). Then let a production subsidy of σ per MWh be paid: how far past the optimum does entry go, and which area in Hirth's figure is the welfare loss?
- A wind developer argues that every new farm lowers the revenue of every existing one, that the market never compensates this, and that wind therefore needs support. Place the argument: which of Hirth's three costs is it, and is it a market failure? Then, against Hirth's sentence: §3.5 gives conditions under which two of his three costs *are* market failures. Which two, under what conditions, and what does the targeting principle prescribe for each?

3. Joskow (2011) argues that comparing wind and gas by levelised cost is an error. State his argument in the language of the model: what is the good, and why is a MWh not a MWh? Use the two-hour system of part 1 at $K = 0.5$ and at $K = 3.5$ as the illustration.

Stop point

The free-entry condition, and one paragraph that places the developer's argument and names the model result it rests on.

If you finish early

López Prol et al. (2020) find in Californian data that solar entry *raises* wind's value factor. A wind lobbyist calls this proof that wind and solar are complements. With part 2's decomposition in hand, what do you answer, and what single number would settle it?